Lecture 13, Math 441

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2nd Order Linear IVP

$$\begin{cases} y'' + \frac{1}{x}y' - \frac{9}{x^2}y = 0 \\ y(1) = 0 \\ y'(1) = 1 \end{cases}$$

(Equidimensional)

Try: $y = x^\alpha$; $y'(x) = \alpha\, x^{\alpha-1}$; $y'' + \alpha(\alpha-1)\, x^{\alpha-2}$

Substituting:

$\alpha(\alpha-1)\, x^{\alpha-2} + \alpha\frac{1}{x}x^{\alpha-1} - \frac{9}{x^2}x^\alpha = 0 \;\rightarrow\; \alpha(\alpha-1) + \alpha - 9 = 0$ timestamp: 00:07

So the solution $y(x) = x^{\alpha}$ works if $\alpha = \pm 3$.

Therefore two solutions: $y_1 = x^3$ and $y_2 = x^{-3}$

Combine them linearly: $y = c_1 x^3 + c_2 x^{-3}$

$y'(x) = 3 c_1 x^2 - 3 c_2 x^{-4}$

Apply the initial conditions:

$y(1) = c_1 + c_2 = 0;\ y'(1) = 3 c_1 - 3 c_2 = 1 \ \rightarrow\ c_1 = \frac{1}{6};\ c_2 = -\frac{1}{6}$

The solution to the I.V.P. is $y(x) = \frac{1}{6}\left(x^3 - x^{-3}\right)$

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Note: if the initial conditions were

$y(1) = c_1 + c_2 = \gamma;\ y'(1) = 3\,c_1 - 3\,c_2 = \delta$, we could still solve for c_1 and c_2, for any value of γ and δ and find a unique solution to the I.V.P.

Consider instead the I.V.P.:

$y'' + y = 0; \qquad y(0) = \gamma;\ y'(0) = \delta$

Two solutions of the equations are $y_1(x) = \cos(x)$ and

$y_2(x) = \mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x}$

Let's combine them: $y(x) = c_1 \cos(x) + c_2 \left(e^{ix} + e^{-ix}\right)$

and $y'(x) = -c_1 \sin(x) + i\, c_2 \left(e^{ix} - e^{-ix}\right)$

Applying the initial conditions:

$y(0) = c_1 + 2\, c_2 = \gamma$

$y'(0) = 0 = \delta$

This can be solved only if $\delta = 0$, not for any value of γ and δ.

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Problem: $\cos(x)$ and $e^{ix} + e^{-ix}$ are not really different from each other, in fact $\cos(x) = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$, so when we combine them we don't really get two degrees of freedom to match the two initial conditions.

Linear Independence

Two solutions are linearly independent if neither is a constant multiple of the other. Otherwise, they are linearly dependent.

Ex: $\cos(x), \sin(x)$ are linearly independent. $1, x$ are LI. $\mathrm{e}^{x}, \mathrm{e}^{2x}$ are LI. $1+x, -2-2x$ are LD. $\sin(x), \frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{2\mathrm{i}}$ are LD. $x-1, x+1$ are LI. $0, f(x)$ are LD for any $f(x)$ because 0 is a constant multiple of any $f(x)$.

Assume $y_1(x)$ and $y_2(x)$ are two solutions of the equation. We form a general solution as $y(x) = c_1\, y_1(x) + c_2\, y_2(x)$ and

$y'(x) = c_1\, y_1'(x) + c_2\, y_2'(x)$

Apply the initial conditions:

$y(x_0) = c_1\, y_1(x_0) + c_2\, y_2(x_0) = \gamma$

$y'(x_0) = c_1\, y_1'(x_0) + c_2\, y_2'(x_0) = \delta$

Wronskian

This system has one and only one solution no matter what γ and δ are as long as

$$\det \begin{pmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{pmatrix} = y_1(x)\, y_2'(x) - y_1'(x)\, y_2(x) \neq 0$$

If y_1 and y_2 are linearly dependent, e.g. $y_1 = \beta y_2$, then

$y_1(x_0)\, y_2'(x_0) - y_1'(x_0)\, y_2(x_0) = 0.$

The determinant above is called the Wronskian of y_1 and y_2, written $W(y_1, y_2)(x)$.

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Abel's Theorem

Assume that y_1 and y_2 are solutions to $y'' + p(x)y' + q(x)y = 0$

So, $y_1'' + p(x)\,y_1' + q(x)\,y_1 = 0$ and $y_2'' + p(x)\,y_2' + q(x)\,y_2 = 0$

Multiply the first equation by y_2 and the second by y_1 and subtract second equation from the first to get

$y_1\,y_2'' - y_2\,y_1'' + p(\boxed{y_1\,y_2' - y_2\,y_1'}) = 0$

Notice the boxed is the Wronskian, $W = y_1\,y_2' - y_2\,y_1'$

$W' = \cancel{y_1'y_2'} + y_1y_2'' - \cancel{y_1'y_2'} - y_1''y_2 = y_1y_2'' - y_1''y_2$. So, we can rewrite

$y_1\,y_2'' - y_2\,y_1'' + p(y_1\,y_2' - y_2\,y_1') = 0$ as $W' + pW = 0$

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This is a separable equation and assuming that $W \neq 0$:

$$\frac{\mathrm{d}W}{W} = -p\,\mathrm{d}x \;\rightarrow\; \ln(W) = -\int^{x} p(s)\mathrm{d}s + A \;\rightarrow\; \boxed{W(x) = C\,\mathrm{e}^{-\int^{x} p(s)\mathrm{d}s}}$$

(N.B., we set $e^{A} = C \neq 0$)

The following is therefore true:

Given two solutions y_1 and y_2 of a linear homogeneous O.D.E. in an interval I, if they are linearly independent, then

$W(y_1, y_2)(x) \neq 0, \ \forall x \in I$ if they are linearly dependent, then

$W(y_1, y_2)(x) = 0, \ \forall x \in I$

This is called Abel's theorem

This also means that to check linear independence we can just calculate the Wronskian at a specific value of $x \in I$.

Ex., consider $y_1 = \cos(x)$ and $y_2 = \left(e^{ix} + e^{-ix}\right)$

Wronskian of y_1 and y_2 is

$$\det \begin{pmatrix} \cos(x) & e^{ix} + e^{-ix} \\ -\sin(x) & i\left(e^{ix} + e^{-ix}\right) \end{pmatrix}$$

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This is difficult to evaluate without remembering Euler's formulas.

But if we calculate this for $x = 0$ we get

$$\det \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = 0$$

So, y_1 and y_2 are linearly dependent.